

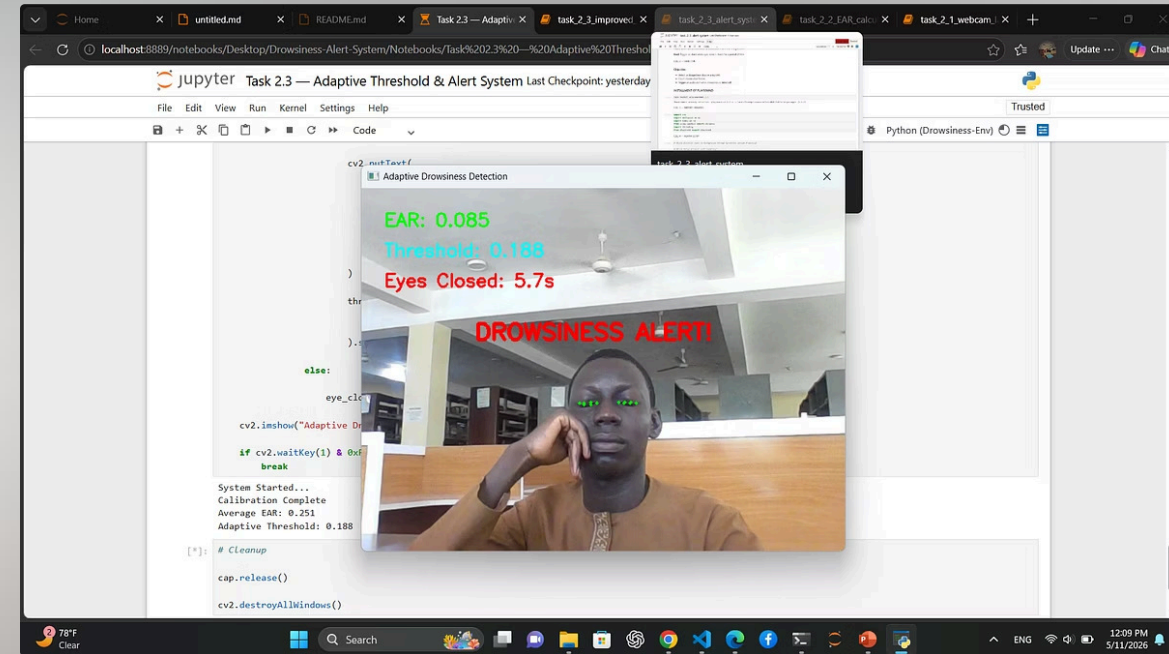
AI Drowsiness Alert System

Real-Time Driver Fatigue Detection Using Computer Vision

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Introduction



Driver fatigue and micro-sleep are major causes of road accidents worldwide. Many accidents occur because drivers become sleepy without realizing it.

This project was developed to create a real-time system that monitors eye behavior and detects signs of drowsiness using Artificial Intelligence and Computer Vision.

Situation

Many existing road accidents are caused by related factors:



Driver Fatigue

Extended periods behind the wheel without rest



Long Driving Hours

Extended time on the road without breaks



Lack of Sleep

Insufficient rest before driving



Micro-Sleep

Brief unconscious episodes while driving

There was a critical need for a system that can monitor the driver's face in real-time and warn the driver before falling asleep.

Task

The main goal of this project was to develop a comprehensive drowsiness detection system:

01

Access Live Webcam

Capture real-time video feed from camera

02

Detect Facial Landmarks

Identify key facial features in real time

03

Isolate Eye Regions

Focus on eye area detection and monitoring

04

Calculate EAR

Compute Eye Aspect Ratio values

05

Differentiate States

Separate blinking from actual drowsiness

06

Trigger Alarm

Activate alert during prolonged eye closure

✅ **Additional Features Implemented:** Timer-based detection, adaptive thresholding, and real-time alert system

Technologies Used

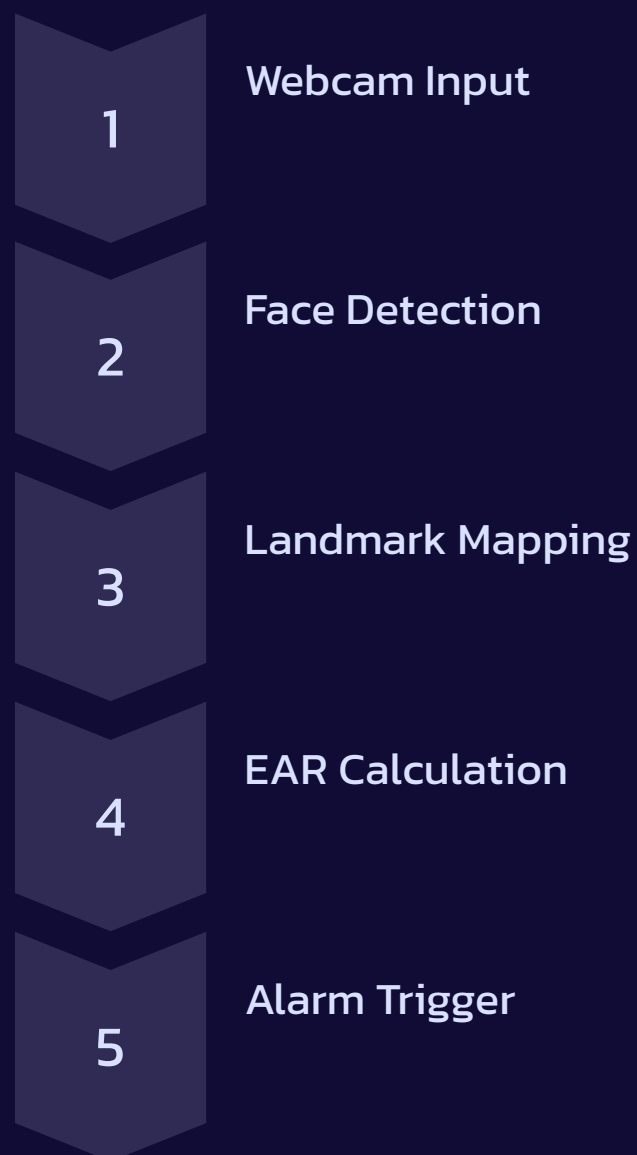
Programming & Libraries

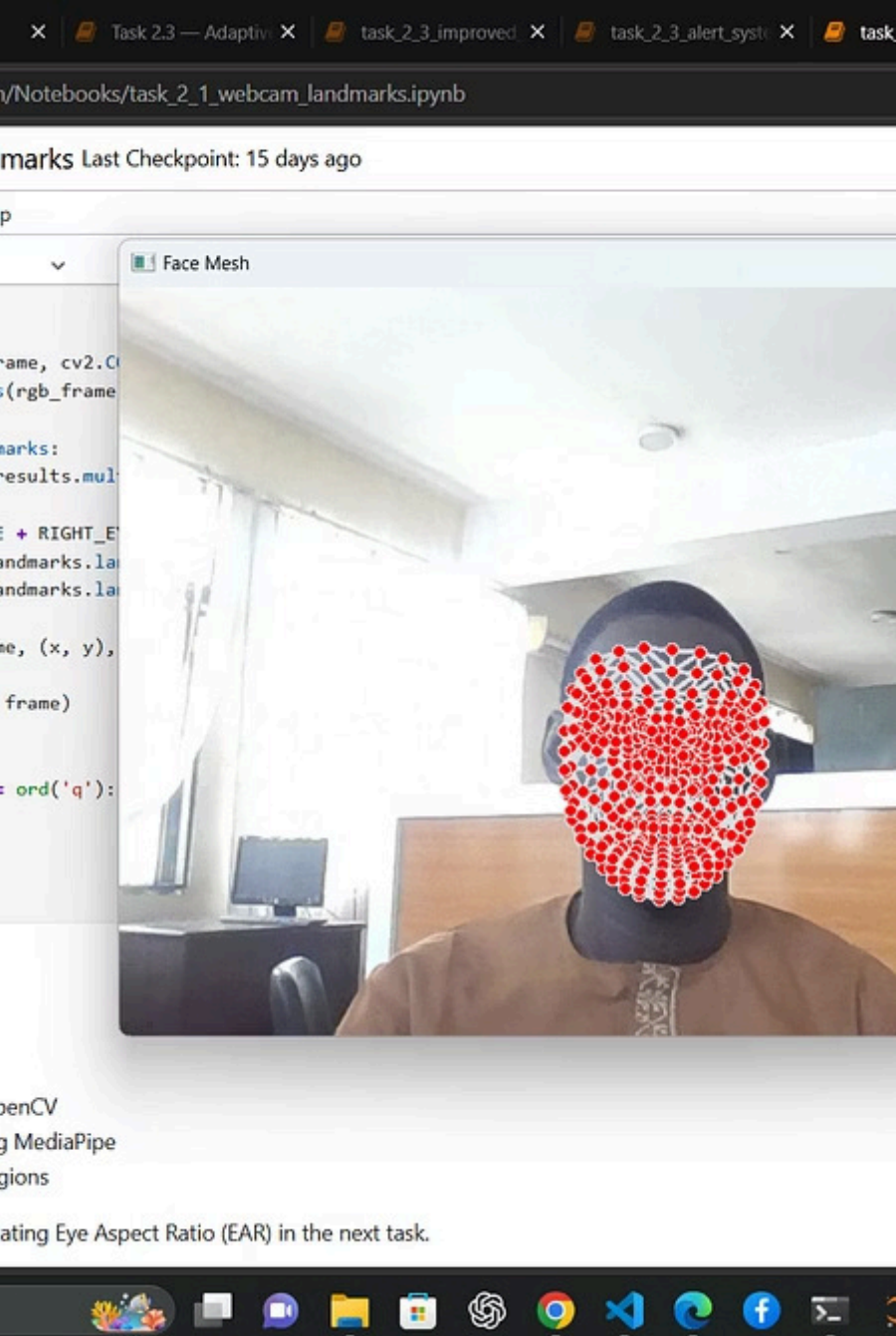
- **Python** - Core programming language
- **OpenCV** - Computer vision processing
- **MediaPipe** - Facial landmark detection
- **NumPy & SciPy** - Mathematical computations
- **Playsound** - Audio alert system

Key Concepts Applied

- Computer Vision
- Facial Landmark Detection
- Feature Extraction
- Real-Time Video Processing
- Eye Aspect Ratio (EAR) calculation

System Workflow





Action: Facial Landmark Detection

The system uses OpenCV for webcam access and MediaPipe Face Mesh for precise facial landmark detection.

468 Facial Landmarks

Comprehensive facial feature detection across entire face

Eye Landmark Coordinates

Real-time tracking of eye position and movement

Continuous Monitoring

Webcam tracks user's eyes and facial movement constantly

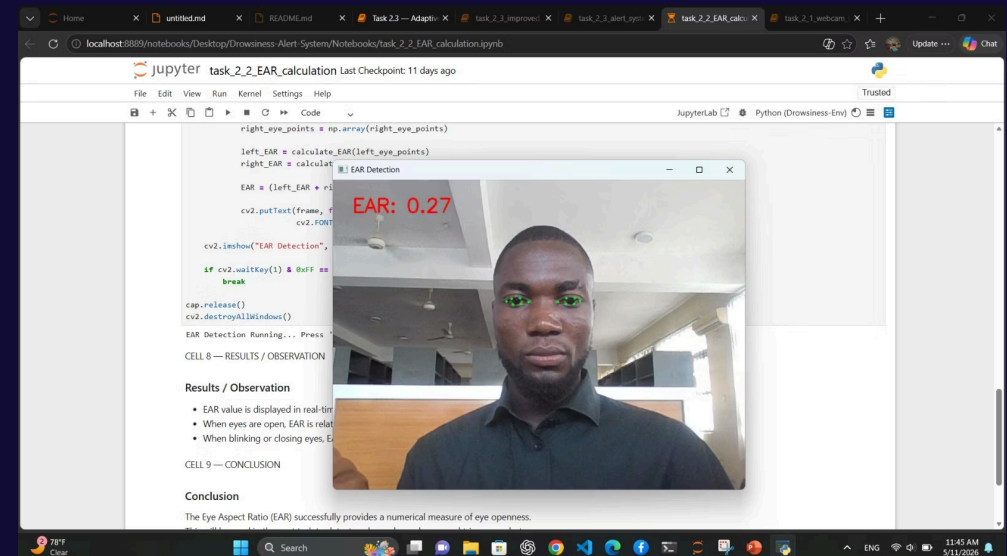
EAR Calculation

The Eye Aspect Ratio (EAR) determines whether the eyes are open or closed:

EAR Formula Interpretation

- **Higher EAR** → Eyes Open
- **Lower EAR** → Eyes Closed

The EAR value changes dynamically during blinking and sleeping states.



EAR Interpretation

👁️ Eyes Open

Higher EAR values indicate open eyes

👁️ Blinking

Temporary EAR drop during normal blink

😴 Eyes Closed

EAR drops below threshold during drowsiness

Adaptive Threshold & Timer Logic

To reduce false alarms, the system implements intelligent detection mechanisms:



Observe Normal State

System first observes user's normal eye state and behavior



Adaptive Calibration

Automatically adjusts EAR threshold for different eye shapes



Timer Validation

Checks duration of eye closure continuously



Decision Logic

Short closure → Blink | Long closure (4+ seconds) → Drowsiness

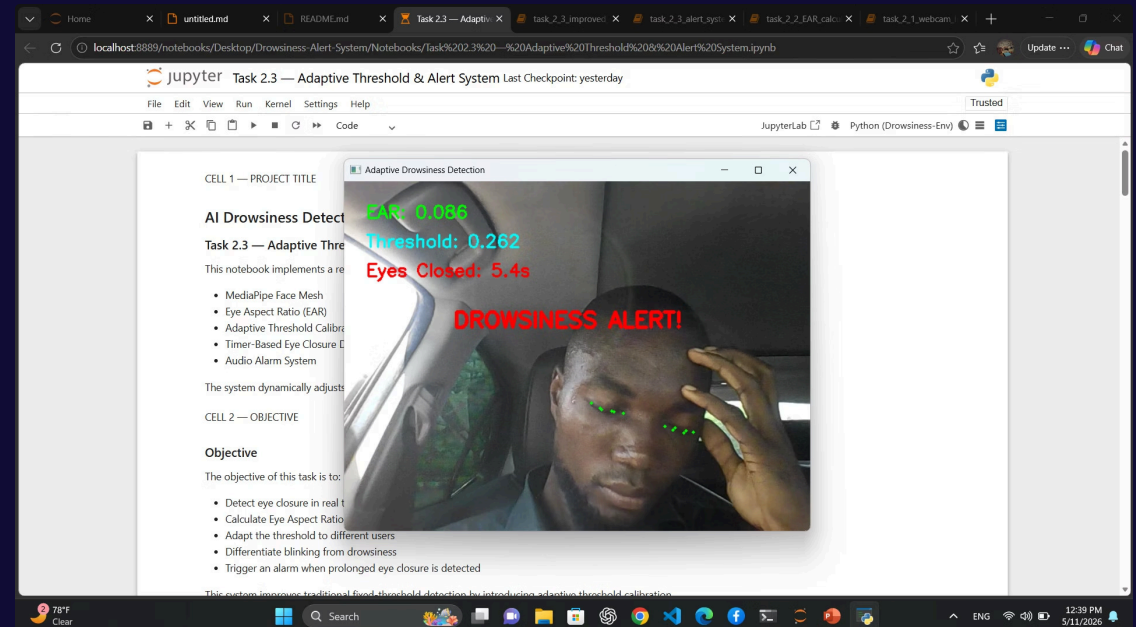
i This approach significantly improved accuracy and reduced false positives in real-world testing.

Results & Future Improvements

Project Outcome

- ✓ Successfully detected eye landmarks in real time
- ✓ EAR values updated dynamically
- ✓ Differentiated blinking from drowsiness
- ✓ Triggered audio alarm during prolonged eye closure
- ✓ Worked successfully on standard laptop hardware

Final Output: The system can monitor a driver and warn them before falling asleep.



Limitations & Future Improvements

Current Limitations

- Focuses mainly on eye behavior
- Does not yet detect yawning
- No head pose estimation
- Performance may reduce under poor lighting
- Glasses or occlusion may reduce reliability

Future Enhancements

- Head pose estimation
- Yawning detection
- Deep learning-based classification
- Mobile deployment
- Edge AI deployment (Raspberry Pi / Jetson Nano)
- Smart vehicle dashboard integration

Real-World Applications & Conclusion

This project belongs to the fields of Artificial Intelligence (AI), Computer Vision, Human Monitoring Systems, and Embedded Safety Systems.



Smart Vehicles

Integration into modern automobile safety systems



Driver Assistance

Real-time monitoring and alert systems



Transportation Safety

Commercial fleet and logistics monitoring



Industrial Monitoring

Machine operator fatigue detection



Edge AI Cameras

Embedded vision systems for safety



Cloud Monitoring

Centralized surveillance and analytics

Conclusion: This project demonstrates how Artificial Intelligence and Computer Vision can improve safety by detecting driver drowsiness in real time. The system provides a foundation for future intelligent driver-assistance technologies.